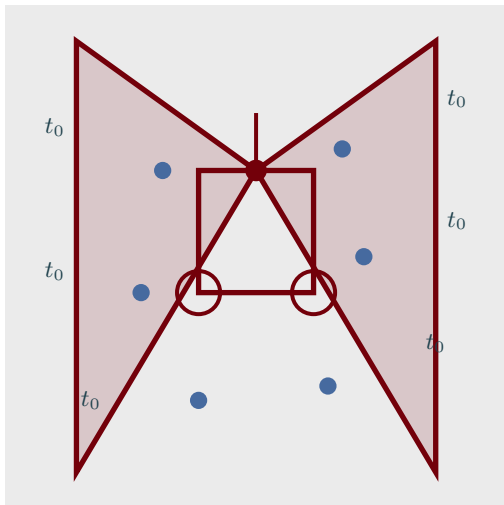


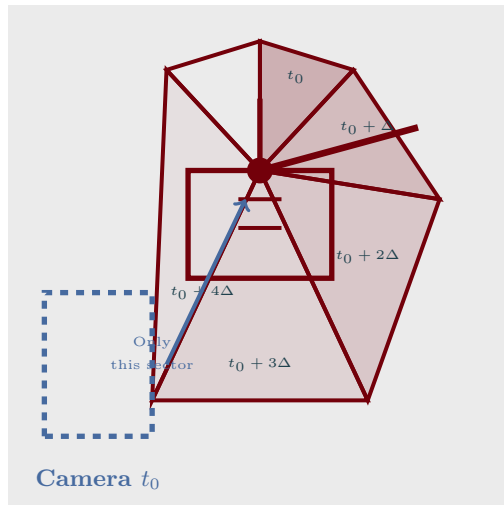
Automotive Radar



(Electronic Steering)

All targets measured at
same time t_0

Marine Radar



(Mechanical Rotation)

Different azimuths measured
at different times

1-3 sec full rotation
→ bearing-dependent timing error

Temporal
Mismatch

Figure 1: Temporal synchronization challenge for mechanically rotating marine radar. Unlike electronically steered automotive radar that illuminates the entire FOV quasi-simultaneously, a rotating marine radar antenna measures different azimuths at different times within each sweep, creating bearing-dependent timing errors when associating with camera frames.